Humanities & Social Sciences Communications

ARTICLE

https://doi.org/10.1057/s41599-025-06265-x OPEN

Improving students' reading, comprehension, and reading motivation through texts generated from their prompts using ChatGPT-4: a mixed-method study

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Reading enables individuals to acquire knowledge, think critically, succeed in social life, access new information, and remain continually informed. This underscores the need to emphasize reading education in the elementary years when the foundation of literacy is established. This study aims to improve the reading, comprehension, and motivation of 4th-grade students through ChatGPT-4, which generates texts based on their own prompts. The study was conducted in two public primary schools located in the center of Bursa, Türkiye. The research group consisted of 106 students. The study employed a sequential explanatory mixed method design. The quantitative aspect of the study was conducted using the Solomon four-group experimental design, while the qualitative part was carried out through a case study. Quantitative data were analyzed using the Mann-Whitney *U* test, Independent Samples *t*-test, ANOVA, and effect size calculations, while qualitative data were examined using descriptive and content analysis. According to the findings, when the experimental groups were compared with the control groups, statistically significant differences were identified in favor of both experimental groups with respect to reading and comprehension skills as well as reading motivation. These differences showed a high effect size across all dependent variables. Based on these findings, it is recommended that teachers integrate ChatGPT to help students generate appropriate texts during free reading periods.

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Introduction

Thanks to rapid advancements in information technologies, artificial intelligence applications capable of mimicking human behavior are now being utilized across various fields (Jia, 2023). Due to these developments, ChatGPT, one of the artificial intelligence applications, is increasingly being used to enhance educational environments (Elmas et al., 2024). This has led to a growing interest in research focused on determining ChatGPT's contributions to students' learning processes (Mahapatra, 2024; Woo et al., 2024). One of the areas these studies focus on is the development of students' reading skills and reading motivation. The significance of such studies stems from the profound impact of reading on an individual's personal and academic life. A positive relationship has long been established between the development of students' reading skills and their academic success (Anderson et al., 1988). Moreover, this relationship fosters the growth of cognitive skills that are essential for sustained academic achievement (James et al., 2024). Reading and comprehension are not only essential for students' academic success but also fundamental competencies they will rely on throughout their lives (Moss et al., 2024). Moreover, proficiency in reading and comprehension plays a crucial role in achieving social success, and fostering these skills is a fundamental responsibility of education(Lorch & Broek, 1997). In this regard, reading and comprehension can be enhanced through a variety of pedagogical interventions (Olimova & Gilyazetdinov, 2024).

Enhancing students' reading skills and motivation has been the focus of numerous studies. In these studies, while traditional teaching methods have been preferred (Fatimah et al., 2024; Ibrahim, 2017; Miller & Meece, 1997; Nevo & Vaknin-Nusbaum, 2020; Notanubun et al., 2024; Permatasari & Wienanda, 2023), technology-supported approaches have also become more widespread in recent years (Anggrainy et al., 2024; Gea et al., 2024; Hong, & Lee, 2023; Hossu, & Roman, 2023). Interactive reading applications, audiobooks, multimedia resources, and artificial intelligence play a crucial role in enhancing reading skills and motivation (Rofi'i et al., 2023). Alongside research on artificial intelligence, the impact of teaching methods on students' reading skills should not be overlooked (McLean, 2007). The integration of technology into instruction has proven effective in developing elementary students' comprehension skills (Chuyen & Dong, 2023). In this regard, the present study distinguishes itself by incorporating AI-supported texts and activities alongside traditional teaching methods. Furthermore, a review of the literature on ChatGPT's role in developing primary school students' reading skills shows that existing studies have focused on areas such as general reviews (Jendia & Ismail, 2023)(Jendia & Ismail, 2023), the readability of ChatGPT's responses to students' questions (Murgia et al., 2023), its integration into primary school learning (Kockova et al., 2023), and its use in teaching foreign languages (Wotsa & Yadav, 2023). This study differs from previous research by focusing on the enhancement of reading, comprehension, and reading motivation in the mother tongue, addressing decoding difficulties, and exemplifying a triadic interaction model through the integration of artificial intelligence into a teaching approach grounded in teacher–student interaction. ChatGPT enables students to take responsibility for their own learning by making them active in the text creation process. In this context, the practice employs a hybrid model in which students' prompts shape the creation of texts and activities for mother tongue education, and these texts are subsequently implemented through established teaching methods. Such an approach is anticipated to be both pioneering and original.

Conceptual framework

Reading and reading motivation. Reading and comprehension, fundamental to communication and the foundation of knowledge acquisition and professional expertise, are complex and transformative processes that shape thinking and perception (Olimova & Gilyazetdinov, 2024). Reading remains central to daily life and plays a vital role in information acquisition (McCarthy & Yan, 2024). Reading is a fundamental pillar of education, enabling students to engage effectively with learning activities, acquire knowledge, analyze and interpret information, and develop critical thinking skills (Madaminova, 2024). Classroom activities, assignments, and all forms of reading materials —books, stories, articles, and chapters—are ultimately rooted in comprehension (Pearson, 2014). Reading is an essential skill for learning, allowing students to access and benefit from educational materials independently. Therefore, if students fail to develop adequate reading skills in early education, they may face difficulties in later stages of schooling and in their subsequent lives (James et al., 2024).

The true measure of success in reading is comprehension (Moss et al., 2024; Paige et al., 2024). According to Kintsch (1979), defining comprehension in all its dimensions is extremely difficult. Kirby (2007) argued that reading comprehension operates in a bottom-up manner, progressing from sentences to details, then to main ideas, and ultimately to overarching themes. With the help of words, more abstract ideas are derived from sentences based on grammatical rules. Understanding at the level of ideas represents a more superficial interpretation of the text. Through interaction, these ideas are synthesized to uncover the main ideas, from which thematic generalizations and abstractions are derived. In reading comprehension, readers construct an abstract model of the situation described, moving beyond the cognitive processing of individual words (Van Dijk & Kintsch, 1983). Successful comprehension is largely dependent on the effective formation of this mental model (Moss et al., 2024). During reading comprehension, individuals perform cognitive tasks that link words, sentences, and paragraphs to construct a coherent mental representation of the text. At the same time, they use metacognitive skills to monitor the coherence of these representations and make necessary adjustments (Soto et al., 2019). These skills enable readers to monitor comprehension by being aware of their cognitive processes during reading and regulating subsequent actions to achieve learning objectives (Schraw & Dennison, 1994).

Collins and Smith (1980) proposed a two-process model that conceptualizes comprehension as an active process. The first component involves monitoring comprehension. In this stage, readers take corrective actions to address potential failures, such as difficulties in understanding relationships between words, sentences, or larger textual units. The second processing skill involves using clues in the text to generate, evaluate, and revise hypotheses about current and future events. At this stage, readers consider variables such as text type, events, and character traits. They make predictions about the progression of the text by forming hypotheses, testing their accuracy, and interpreting its key points. Predictions concern what may occur, whereas comments focus on interpreting what is already present in the text. These two skills are interdependent and closely connected. For instance, temporary hypotheses can be formed regarding the meaning of a word or sentence. Generating a hypothesis or making a prediction may concern the overall text, but it can also focus on the meaning of a word or sentence at the micro level. The main goal of the reading model is to help students identify potential comprehension challenges during silent reading and develop the skills to overcome them. To achieve this, practice sessions are conducted under teacher supervision during read-aloud activities. Over time, the teacher's guidance gradually diminishes, enabling students to apply the model independently

while reading. In the final stage, the goal is for students to apply these skills while reading silently.

According to Gough and Tunmer's (1986) Simple View of Reading, reading comprehension (R) consists of two components—decoding (D) and language comprehension (C)—and is best represented by a multiplicative model ($R = D \times C$). Linguistic comprehension is also referred to as listening comprehension. The multiplicative relationship indicates that comprehension cannot occur without both word recognition and language comprehension. For instance, if $R = D \times C$ and $C = 0$, then R equals 0. Gough and Tunmer illustrate this with the example of a five-year-old child who can use and understand language but cannot read due to underdeveloped word recognition skills. According to this model, comprehension is high only when both word recognition and language comprehension skills are strong. If word recognition is strong but language comprehension is weak, comprehension remains low. Similarly, if language comprehension is strong but word recognition is weak, comprehension also remains low (Georgiou et al., 2009).

Apart from the Simple View of Reading, it can be stated that there are three traditional theories that explain the reading process. The first of these is the Bottom-up Theory, which focuses on the decoding (D) component of the Simple View of Reading. According to this theory, reading begins when the units that constitute the text are received by the reader as visual input (Ada Amadi, 2019). From the smallest units—phonemes to syllables, syllables to words, and words to sentences—meaningful fragments are formed and subsequently integrated to construct overall comprehension (Field, 1999). Each operation performed and the hierarchically structured data serve the purpose of reaching meaning. In this respect, it can be argued that the model explains reading by placing the text at the center. Advocates of this model contend that in order for students to acquire higher-level comprehension skills, they must first develop automaticity in the foundational skills of early literacy (Mondesir & Griffin, 2020).

Another theory is the Top-down Theory, which focuses on the linguistic comprehension (C) component of the Simple View of Reading. This theory emphasizes an interaction between the reader and the text in which the reader's prior knowledge, experiences, predictions about the text, hypotheses, and expectations are actively engaged (Zhang, 2010). In this model, the reader reconstructs meaning by drawing on these expectations regarding the text (Pourmohammadi, 2016). Therefore, unlike the Bottom-up Theory, the Top-down Theory can be said to place the reader at the center.

The third theory is Interactive Reading, which integrates the processes of the Bottom-up and Top-down theories (Stonavich, 1980). Interactive Reading is a model based on the interaction established between students and an adult before, during, and after reading. During the reading process, participants listen to one another, evaluate ideas, and collaboratively engage in activities to read and comprehend the text. In this process, students' interests, needs, and background knowledge about the text play an active role (Ashford, 2024). This process ensures that students remain active both affectively and cognitively. Interactive Reading is grounded in the principles of encouraging the child's active participation in reading, fostering language use, providing feedback on language production, and enabling the adult to adapt their own reading practices to the child's linguistic abilities (Pillinger & Wood, 2013). When implemented in the early years, it facilitates the resolution of potential reading difficulties (Morgan & Meier, 2008). In this study, this reading approach was adopted. In the common practice of Interactive Reading, listening and speaking skills are employed alongside reading skills. Within the scope of the research, since the interaction began during the text-generation process, the adopted approach allowed for the observation, assessment, feedback, and development of all students' language skills.

Hirsch (2003) proposes three principles to enhance students' reading comprehension: fluency, vocabulary, and domain knowledge. Fluency enables the reader to focus on comprehension; a broad vocabulary facilitates understanding and learning; and domain knowledge—also referred to as prior subject knowledge—supports both fluency and vocabulary, thereby fostering deeper comprehension. These three principles are closely interconnected. Fluent students who can pronounce words quickly and accurately are able to master decoding and concentrate on comprehension without overloading their memory. If decoding is slow, previously decoded words may be forgotten during the ongoing process, which can hinder comprehension of the text. Students can overcome the limitations of working memory by reading fluently, recognizing words rapidly, and applying basic grammatical rules. Fluent readers can focus on comprehension because they have automated many of the fundamental processes involved in reading. A broad vocabulary supports fluent reading, as students are already familiar with the words. Domain knowledge, defined as the reader's familiarity with the text's content, also contributes to this process. Students with prior subject knowledge tend to read more fluently on a given topic, as they are familiar with it and have already encountered the relevant vocabulary. Experts in a subject can read texts related to that subject more fluently, as their prior knowledge enables them to process information more efficiently and draw inferences by connecting new material with what they have already learned.

The critical role of vocabulary knowledge in determining reading comprehension success has been widely recognized (Sidek & Rahim 2015). A strong relationship exists between vocabulary richness and reading comprehension (Torgeson et al., 1997). Because word recognition involves repeated cognitive activation during reading, comprehension is considered to depend on successful word decoding (Perfetti, 2007). Verhoeven and van Leeuwe (2008) state that the relationship between vocabulary and comprehension is reciprocal: vocabulary expands through reading, and comprehension improves as vocabulary develops. In proficient readers, decoding and comprehension are closely integrated, and both skills are typically well developed. In contrast, weak readers tend to have difficulties in both decoding and comprehension, with each skill reinforcing the other's weakness (Shankweiler, 1999).

Reading fluency is closely linked to comprehension (Fuchs et al., 2001). Reading fluency is an indicator of advanced reading skills (Reis et al., 2008). Poor fluency leads to slow word recognition. As a result, previously recognized words may be lost from working memory before subsequent words are processed, hindering their integration For readers who struggle with fluency, the first step is accurate reading. However, focusing too much on accurate word pronunciation may hinder overall comprehension. Thus, attention to accuracy at the word level is often associated with inadequate comprehension (Cain, 1999).

Since reading supports both individual and societal development, it is essential for students to cultivate a love of reading and sustain their interest in it (Seventilova, 2024). According to Bartlett (1932), understanding a text involves active and constructive cognitive processes rather than passive and receptive ones. During this process, readers continuously build a mental representation of the text by using their prior knowledge. This active process requires a willingness to engage in reading. Reading motivation influences whether students develop positive or negative attitudes toward reading. Students with high reading motivation exert the necessary effort and use strategies to comprehend the text, as they recognize the challenges of the

reading process (Ahmadi et al., 2013). The effort and persistence needed for comprehension are fueled by reading motivation (Baker & Wigfield, 1999).

As motivation increases, reading and comprehension skills improve accordingly (Larson et al., 2016). When students relate what they read to their own experiences, their motivation to read increases (Briggs, 2010). Texts that are unrelated to students' interests weaken their reading motivation (Guia et al., 2024). Low reading motivation contributes to difficulties such as limited vocabulary, poor decoding skills, lack of fluency, and weak phonological awareness during reading and comprehension (Kheang et al., 2024). Sharing their thoughts and comparing them with others engages students in the learning process and fosters their reading motivation (Wahidah, 2018). Reading motivation can be cultivated through an enjoyable classroom environment, engaging reading activities, and texts aligned with students' interests (Mohseni Takaloo & Ahmadi, 2017).

To foster reading motivation, it is important that students are active in the reading and comprehension process, that texts match their interests, that they can relate what they read to their experiences, and that they are supported in overcoming limited vocabulary. Since ChatGPT offers features that support students in these areas, it has been considered a motivational enhancer in this study.

ChatGPT and its use in the learning environment. Artificial intelligence applications offer significant opportunities to enhance reading and comprehension tasks, which have become increasingly complex in modern life. These technologies support students' reading comprehension by providing materials appropriate to their level (McCarthy & Yan, 2024). Applications accessible through technological tools such as tablets and smartphones can help students with reading difficulties improve their skills (Sand et al., 2024).

In addition, multiple Large Language Models (LLMs) were utilized with respect to students' reading skills. Bidirectional Encoder Representations from Transformers (BERT) was employed to address students' attention difficulties during reading and to analyze reading comprehension questions (Chen & Wu, 2020; Fernandez et al., 2022); The Text-to-Text Transfer Transformer (T5) was utilized to provide a reading comprehension model, generate questions, and produce answers (Guan et al., 2024; Guo et al., 2022); The Cross-lingual Language Model–RoBERTa (XLM-R) was applied to perform multilingual inferential reading and comprehension tasks across different languages (Choi et al., 2021; Wu et al., 2021) Beyond these models, the Chat Generative Pre-Trained Transformer (ChatGPT), developed by OpenAI, has been widely adopted in the enhancement of reading skills.

ChatGPT is an artificial intelligence–based application designed to generate text in response to user prompts (Halaweh, 2023). ChatGPT is a large language model that improves its outputs through deep learning. It draws on a vast repository of texts and images from the internet to generate responses to user prompts. With the support of DALL-E, it allows individuals in interactive learning environments to create visuals tailored to their preferences (Rospigliosi, 2023). While some perspectives encourage the use of ChatGPT, others emphasize its potential negative impacts (Adeshola & Adepoju, 2023). Halaweh (2023) describes these concerns as issues that accompany every technological development. Despite differing opinions, ChatGPT has found applications in education. The use of ChatGPT and similar technologies in education has become an unavoidable reality (Loos et al., 2023).

ChatGPT, which can generate human-like texts, differs from other language models by focusing on how people express themselves, aiming to establish more natural communication and meet their needs (Yusuf, 2023). Susnjak and McIntosh (2024) examined ChatGPT's ability to perform high-level cognitive tasks and generate human-like text, noting that it can apply critical thinking skills and produce highly realistic output with minimal input. As a continuously developing model, ChatGPT allows updates to be made to the generated texts (Brown et al., 2020).

For students to create a mental representation of a text during comprehension, their prior knowledge of the underlying structures must be sufficient (Eleman & Oslund, 2019). Based on this, students whose comprehension skills are below expectations can improve by strengthening their prior knowledge. ChatGPT enables students to supplement missing prior knowledge in the generated texts. Students can address these gaps by asking ChatGPT questions in areas where their prior knowledge is insufficient. Furthermore, responsibility for the comprehensibility of a text lies primarily with the author (Adams, & Bruce, 1982). In texts generated with ChatGPT, students who take responsibility for composing the text also review it for language and narrative features to improve overall quality. According to Gunter and Kenny (2008), discussions among readers can encourage reflection on the events and concepts presented in a text. In this context, ChatGPT–student–teacher interaction enables discussion and reflection on the text by drawing insights from three sources. Thus, discussion and reflection on the text become possible through insights contributed by ChatGPT, students, and teachers.

Murgia, Pera, Landoni, and Huibers (2023) evaluated ChatGPT's performance based on responses to online questions from 9–10-year-old fourth-grade students with different literacy levels, in relation to the elementary school curriculum. Analysis of student feedback revealed that ChatGPT's responses were appropriate for the fourth-grade level. Hidayat (2024) recommends that educators and administrators use AI-based applications to improve students' reading skills, based on his findings. Xiao, Xu, Zhang, Wang, and Xia (2023) examined the use of ChatGPT to create reading materials and comprehension activities for facilitating English learning among middle school students. The study found that ChatGPT generated texts more effective than those created by humans. Bezirhan and von Davier (2023) tested the appropriateness and readability of texts generated by ChatGPT-3 for fourth-grade students. Experts reviewed the generated texts and concluded that they were suitable for the students' level. Additionally, Lin and Chen (2024) showed that, beyond text generation, ChatGPT can produce multiple-choice tests comparable to human-made ones for developing and assessing students' comprehension skills. Findings from the literature confirming ChatGPT's potential to generate questions for text creation and comprehension assessment serve as the primary motivation for this study. As an advanced AI tool, ChatGPT possesses the ability to generate prompt-based texts, revise generated content in line with new prompts, and even create images that reflect the text. These features made it a suitable choice for the present study, as they directly support the research aims. Considering these data along with Alshater's (2022) view that ChatGPT has the potential to support academic studies, this study was conducted using ChatGPT-4.

In this study, texts generated with ChatGPT-4 based on the prompts of 4th-grade students, together with related questions and visuals, are intended to enhance students' reading, comprehension, and reading motivation. In line with this purpose, the following research questions were addressed:

Do the texts generated with ChatGPT-4, based on prompts from 4th-grade students, improve their reading skills?

Do the texts generated with ChatGPT-4, based on prompts from 4th-grade students, improve their comprehension skills?

Do the texts generated with ChatGPT-4, based on prompts from 4th-grade students, improve their reading motivation?

Method

Study model. This study employed a mixed-method approach, defined as a technique that integrates quantitative and qualitative research methods (Johnson & Onwuegbuzie, 2004), together with Creswell's sequential-explanatory design (QUAN→qual). In this design, the researcher first collects and analyzes quantitative data, followed by qualitative data collection to gain deeper insights (Creswell & Clark, 2015). Related quantitative and qualitative data are integrated in the discussion and interpretation sections. The core principle of this design is that combining quantitative and qualitative approaches is more effective for problem-solving than using either approach alone (Creswell, 2014).

The quantitative component of the mixed-method design employed the Solomon four-group experimental design. This design is an approach that aims to address the pre-test sensitivity issue while retaining the advantages of conducting a pre-test, by randomly assigning participants to either receive or not receive the pre-test and intervention program (Gagnon & Barber, 2018). The design consists of four procedural stages. First, four groups are formed. Second, two groups receive a pre-test. Third, the intervention program is applied to one pre-tested group and one non-pre-tested group. Finally, all four groups take a post-test (Huck & Sandler, 1973). In this context, four 4th-grade classes, as detailed in the next section, were selected. The groups to be tested or not tested, as well as the classes designated to receive the pre-test, were randomly assigned, and the pre-tests were administered. Following the implementation of the experimental design, a post-test was administered to all classes.

The qualitative component of the study employed a holistic single-case study design. A case study is a research method that provides a detailed understanding of a situation, or in other words, a case bounded by specific spatial and temporal conditions (Gustafsson, 2017). With this approach, real-life problems can be explored in depth (Ratnasari & Sudradjat, 2023). The holistic single-case design complements traditional research by capturing phenomena and details within a single case (Zaytseva & Bao, 2015). In this context, the interaction between the students participating in the experimental application and ChatGPT-4 was treated as a single case. Through interviews with students, this interaction was explored from multiple perspectives.

The use of ChatGPT not only enhances students' reading comprehension by encouraging their active participation in the instructional process (El Hassan & Alsalwah, 2025) but also enriches their vocabulary (Young & Shishido, 2023). In addition, ChatGPT is capable of analyzing user-generated prompt sentences, deciphering their semantic structures, and producing appropriate texts accordingly (Hadi et al., 2023). Due to these features, ChatGPT was incorporated into the intervention program within the research model.

The Solomon four-group experimental design employed in the study is presented below in Fig. 1.

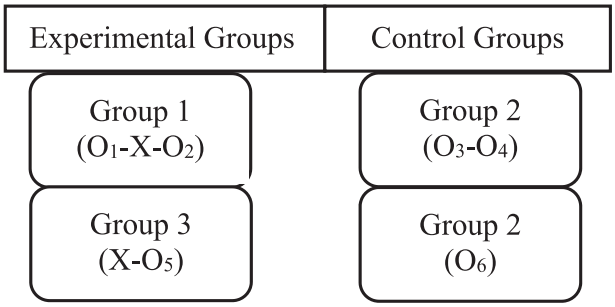


Fig. 1 Solomon four-group experimental design. This figure shows the Solomon four-group experimental design, which includes two experimental groups and two control groups. In the figure, the symbol 'O' represents the measurements taken before and after the experimental procedure, and the symbol 'X' represents the experimental procedure.

Study group. In this study, the sequential mixed-methods sampling design (quantitative–qualitative), commonly used in mixed-methods research, was employed (Teddlie & Yu, 2007). In this design, the quantitative study group is selected randomly, whereas the qualitative study group is determined through purposive sampling (Creswell, 2021). Based on the data collected in the first phase, additional data are gathered from participants in the second phase to provide accurate and reliable information about the study results (Driscoll et al., 2007). Following this approach, two primary schools in the central district of Bursa, Türkiye, were selected through random sampling. Based on the data obtained at the end of the experimental study and the study's objectives, a subgroup of experimental students was selected to collect qualitative data.

The characteristics of the study groups, the data collection tools, and the procedures for data collection and analysis are presented separately for the quantitative and qualitative components.

Characteristics of the quantitative study group. The study group consisted of students from four randomly selected 4th-grade classes at Koç Primary School and Müşerref-Muzaffer Samda Primary School, located in the central district of Bursa, Türkiye. Since the implementation was carried out at the class level, only the sections, and within them the experimental and control groups, could be randomly assigned. All students within these classes were included in the study. According to the Turkish Statistical Institute (2023), the schools where the study was conducted are located in residential areas classified as C2—lower-middle socioeconomic level—with a score of 129. In this respect, no difference exists between the schools. At Koç Primary School, the class in which the experimental application was conducted did not receive a pre-test [Experiment 1 (E1) – 27 students], whereas the class without the application received a pre-test [Control 1 (C1) – 29 students]. At Müşerref-Muzaffer Samda Primary School, the class in which the experimental application was conducted received a pre-test [Experiment 2 (E2) – 25 students], whereas the class without the application did not receive a pre-test [Control 2 (C2) – 24 students]. To determine the sample size, F tests and a priori power analyses were conducted using the G*Power 3.1.9.7 program. In a meta-analysis of the effect sizes of reading studies by Flynn et al. (2012), the mean effect size of 70 was calculated as 0.41. Based on this data, the effect size was accepted as 0.40 in the power analysis, and the test was conducted based on a significance level of .05 and 80% power ($1-\beta = 0.80$). The test concluded that a total sample size of 76 individuals for four groups was sufficient. Therefore, the study group of 105 individuals is considered sufficient. Detailed information about the quantitative study group is provided in Table 1.

Examination of Table 1 shows that, in the quantitative study group, 54 students are girls and 51 are boys; 102 students have internet access at home; 97 own personal technological devices; 33 have magazine or newspaper subscriptions; and 82 have home libraries. Moreover, none of the students had prior experience using artificial intelligence.

Quantitative data collection tools

Error analysis inventory. The inventory was developed by Akyol (2019), drawing on the works of Harris and Spay (1990), Ekwall

Table 1 Characteristics of the quantitative study group.

Groups	Gender		Home Internet		Personal technological device		Magazine or newspaper subscription		Personal bookshelf at home	
	Female	Male	Available	None	Available	None	Available	None	Available	None
E1	13	14	26	1	22	5	7	20	17	10
E2	14	11	23	2	25	-	8	17	18	7
C1	16	13	29	-	28	1	10	19	28	1
C2	11	13	24	-	22	2	8	16	19	5
TOTAL	54	51	102	3	97	8	33	72	82	23

and Shanker (1988), and May (1986), to determine students' reading levels. Using this inventory and its accompanying tables, students' word recognition and comprehension percentages can be calculated. Three levels were defined to interpret the percentages: the *Free Level*, referring to the student's ability to read and understand the text without support; the *Instructional Level*, referring to the student's ability to read and comprehend the text adequately with the support of a teacher or adult; and the *Frustration Level*, referring to the student's limited understanding of the text and frequent reading errors. These categories make it possible to determine the reading levels of elementary and middle school students".

When applying the inventory, it is recommended that the text contain 25–100 words for first- and second-graders; 100–200 words for third- and fourth-graders; 200–300 words for fifth- and sixth-graders; and 300–350 words for seventh- and eighth-graders. It is also suggested that texts be selected from one grade level below the student's current level and that the student's reading be recorded using a voice recorder. By identifying reading error types—such as addition, repetition, and finger tracking—and the symbols representing them, the usability of the inventory was improved.

The word recognition percentage was determined by matching the total number of words in the text with the number of misread words in the relevant table. For example, if a student read a 192-word text with five errors, the table entry corresponding to 191–195 words with five errors indicates a word recognition rate of 97%. In order for a reader to be considered at the independent reading level, the word recognition percentage must be at least 99%. Readers whose recognition percentage falls below 90% are classified as being at the frustration level.

Students' comprehension levels can be assessed through multiple-choice or open-ended questions. In this study, open-ended questions were preferred, and students' comprehension was measured using six questions: three requiring literal understanding and three requiring deep understanding. The literal comprehension questions were scored as follows: two points for a complete response, one point for a partial response, and zero points for no response. The deep comprehension questions were scored as follows: three points for a complete and effective response, two points for expressing more than half of the expected response, one point for a partial response, and zero points for no response. A student's comprehension percentage was calculated by dividing the score obtained by the maximum possible score if all questions were answered correctly. For example, a student assessed with six open-ended questions can obtain a maximum of 15 points—six points from the three literal comprehension questions and nine points from the three deep comprehension questions. I f this student provides a complete response to one literal comprehension question (two points), a partial response to another literal comprehension question (one point), and complete responses to two deep comprehension questions (six points), while leaving the remaining questions unanswered, the student receives a total of nine points. The student's score (9) is then divided by the maximum possible score (15), resulting in a comprehension level of 60%.

During the administration of the inventory, four unfamiliar texts were used to determine students' reading errors and comprehension scores in the pre-test and post-test. All texts were selected from textbooks and supplementary materials prepared by the Ministry of National Education. In the pre-test, the text Konuşan Kitap (192 words) was used to identify oral reading errors, while the text Mezgit Mehmet (209 words) was used to assess comprehension scores. In the post-test, the text Vecihi Hürkuş (192 words) was used to identify oral reading errors, while the text Hep Büyük Efsanesi (213 words) was used to assess comprehension scores. Examples of the questions included in the texts are: "*Why was the man who grew grapes referred to as the one who produced the largest grapes?*" and "*What is the main idea of the text titled 'The Legend of Hep Büyük'?*"

To determine the readability level of the Turkish texts, the formula proposed by Ateşman (1997) was applied. The calculated readability scores are interpreted as follows: 90–100 = very easy, 70–89 = easy, 50–69 = moderately difficult, 30–49 = difficult, and 1–29 = very difficult.

$$\text{Readability Score} = 198.825 - (40.175 \times \text{average syllables}) - (2.610 \times \text{average words})$$

Accordingly, the texts used in the study were evaluated as follows: Konuşan Kitap (192 words; 20 sentences; 504 syllables) had a readability score of 68.31; Mezgit Mehmet (209 words; 25 sentences; 607 syllables) had a score of 60.49; Vecihi Hürkuş (192 words; 23 sentences; 517 syllables) had a score of 68.88; and Hep Büyük Efsanesi (213 words; 19 sentences; 537 syllables) had a score of 68.33. All four texts fall into the "moderately difficult" category, indicating a consistent difficulty level across the materials used in the research.

Reading motivation scale. The scale developed by Wigfield and Guthrie (1997) and later revised by Wang and Guthrie (2004) was adapted into Turkish by Yıldız (2010). The construct validity of the scale was tested through confirmatory factor analysis. The factor–item correlation coefficients ranged from .41 to .73, and the chi-square fit index ($\chi^2/df = 2.51$) indicated an acceptable level of model fit. The scale comprises two main factors: *Intrinsic Motivation*, including the sub-dimensions of *interest* and *curiosity*, and *Extrinsic Motivation*, which encompasses *recognition, social, competition*, and *compliance*. The reliability of the scale was examined using internal consistency coefficients and the test–retest method. Cronbach's alpha coefficients were calculated as follows: *curiosity*, $\alpha = 0.59$; *interest*, $\alpha = 0.68$; *recognition*, $\alpha = 0.52$; *social*, $\alpha = 0.62$; *competition*, $\alpha = 0.62$; *compliance*, $\alpha = 0.54$; *Intrinsic Motivation*, $\alpha = 0.68$; *Extrinsic Motivation*, $\alpha = 0.82$; and the overall scale, $\alpha = 0.86$. Test–retest reliability was assessed with 71 students 3 weeks after the main administration.

Table 2 Analyses of quantitative data.

	Skewness	Kurtosis	Analyses
Number of pre-test reading errors	1.348	2.439	Mann–Whitney U-Test
Pre-test comprehension scores	-0.328	-0.925	Unrelated Samples *t*-Test
Pre-test motivation scores	-0.234	-0.294	Unrelated Samples *t*-Test
Number of post-test reading errors	1.202	1.327	ANOVA
Post-test comprehension scores	-0.098	-1.086	ANOVA
Post-test motivation scores	0.692	0.382	ANOVA

Table 3 Characteristics of the qualitative study group.

	Gender		Home Internet		Personal technological device		Magazine or newspaper subscription		Personal bookshelf at home	
	Female	Male	Available	None	Available	None	Available	None	Available	None
E1	4	6	10	-	9	1	3	7	6	4
E2	6	4	9	1	9	1	4	6	7	3
TOTAL	10	10	19	1	18	2	7	13	13	7

The results indicated moderate to high correlations between the two applications, with consistency coefficients ranging from $r = .65$ to $r = .81$. The scale is a 4-point Likert-type instrument, scored as follows: 1 = very different from me, 2 = somewhat different from me, 3 = somewhat like me, and 4 = very much like me. It consists of 21 items. Scores on the scale range from a minimum of 21 to a maximum of 84. The Cronbach's alpha coefficient for the scale in the study group was calculated as .800 for the pre-test and .837 for the post-test.

Collection and analysis of quantitative data. The Error Analysis Inventory and the Reading Motivation Scale were administered by the researcher as a pre-test to the E2 and C1 groups and as a post-test to all groups. The reading-aloud section of the Error Analysis Inventory, administered in two stages—first, students read a text aloud, and second, they answered questions prepared for a different text—was individually conducted in classrooms allocated by school administrations for this purpose. The second stage was carried out collectively in the classroom environment. The Reading Motivation Scale was also administered collectively in the classroom setting. During the administration of the scale, the original procedure was followed, with each item read aloud one by one. After marking their responses, students proceeded to the next item. Analyses conducted with the SPSS 25 software package are presented in Table 2 below.

Before selecting the analyses to be conducted, the data were examined to determine whether they followed a normal distribution. According to Tabachnick and Fidell (2013), data can be considered normally distributed when skewness and kurtosis values fall between −1.5 and +1.5. Based on this criterion, all data except the number of pre-test reading errors were normally distributed. Non-parametric tests, including the Mann–Whitney *U* test, were applied to this data group, whereas the Independent Samples *t*-test and ANOVA were performed for the other data sets. To identify the groups responsible for the statistically significant differences in the ANOVA results, Levene's test was examined, and the Games–Howell post hoc test was conducted. The effect size of the significant differences was calculated using *eta squared* (η^2).

Characteristics of the qualitative study group. After the experimental procedure was completed, the collected quantitative data were analyzed, and students with low and high performance on the dependent variables were identified in the E1 and E2 groups. Interviews were conducted with 20 students in total, 10 from each group, who exhibited these characteristics. In addition, to capture the perspectives of students who were both highly and minimally influenced by the ChatGPT-supported intervention, this approach was adopted in participant selection. This strategy was intended to ensure that the results would be more objective and representative of a broader range of student experiences. Detailed information about the qualitative study group is provided in Table 3.

As shown in Table 3, the qualitative study group consists of 10 girls and 10 boys; 19 students have internet access at home; 18 own personal technological devices; 7 hold magazine or newspaper subscriptions; and 13 have home libraries.

Qualitative data collection tool

Interview form. The form was developed to elicit the opinions of participating students regarding the texts and activities generated with ChatGPT-4. In preparing the form, relevant literature was consulted (Aktay et al., 2023; Baidoo-Anu & Ansah, 2023). GPT-4?" Example items include: "What are your thoughts on the reading texts, activities, and visuals created with ChatGPT-4?" and "In what aspects do you think ChatGPT-4 is inadequate when generating texts, questions, and related visuals?" The interview form consists of three main questions, as illustrated above, along with four probing questions designed to elicit more detailed information. The interview questions were reviewed by two field experts; however, no revisions were suggested.

Collection and analysis of qualitative data. The researcher conducted individual face-to-face interviews with the students in classrooms designated by the school administrations, and all sessions were audio-recorded. Parental and school administrative consent for audio recording was obtained. The longest interview lasted 10 min and 48 s, whereas the shortest lasted 2 min and 40 s. Subsequently, these recordings were transcribed. The transcribed data were analyzed using descriptive and content analysis methods. The analyzed data were coded for the students and presented in tabular form. Students were assigned identification codes such as E1S1, E1S2, E2S1, and E2S2. In the coding process, "E1" refers to the first experimental group, "E2" to the second experimental group, and "S1" to the first student. For example, the code E1S1 refers to the first student in the first experimental group, whereas

E2S1 refers to the first student in the second experimental group. The coding was carried out by the researcher. The generated codes were organized under various themes through descriptive analysis and further examined through content analysis, whereby the themes were interrelated to derive more detailed insights.

In qualitative research, validity concerns the accuracy of the results, whereas reliability relates to the replicability of the findings. For validity, the analytical approach employed must be sufficient to uncover the underlying reality, while for reliability, it is necessary that the findings be replicable and that different researchers can measure the phenomena in the same manner. To achieve this, the research process must be presented in an explicit and transparent manner (Yıldırım & Şimşek, 2008). To meet these requirements, the interviews were audio-recorded, and the transcriptions were carried out without error and checked multiple times for accuracy. The generated codes were confirmed by asking the students directly. Detailed explanations were provided regarding data collection and analysis, the procedures followed, the characteristics of the participants, the school environment, the technologies employed, and the implementation process. The qualitative data were presented in tables, where student codes indicated which participant made a particular statement. Additionally, excerpts from participants' statements were placed beneath the tables to allow readers to verify the coding.

Experimental application. Before the experimental application, official procedures—including obtaining permission to use the scale and approval from the ethics committee—were completed. School administrations and teachers were informed about the study, after which the experimental and control groups were visited to explain the activities and tests to the students. Parental consent forms were collected via the students. The measurement tools and the reading texts required for their use were prepared. The Error Analysis Inventory and the Reading Motivation Scale were administered to the pre-test groups (E2 and C1). During the preliminary measurements, school visits were made on consecutive days to accommodate students who were absent. All preliminary measurements were completed within five working days. After these procedures, a 6 h pilot study was carried out with both experimental groups. In the first 2 h of the pilot study, unstructured activities were carried out to help students become familiar with ChatGPT-4. In these activities, students mainly tested ChatGPT-4's accuracy by asking various questions. They asked questions such as "What is the capital of Türkiye?" and "What is İnegöl famous for?"—questions whose answers they already knew from their own experiences or close circles. With the researcher's guidance, the students realized that ChatGPT-4 could generate reading texts. In the two class periods following the free activities, texts, comprehension questions, and visuals were created based on the students' prompts. After the text was completed, the researcher left the school, and the text, questions, and visuals were printed in color. The reading and comprehension activities were conducted during two class periods the following day. The procedures mentioned for the pilot application were carried out in both experimental groups. The printed materials were well received by the students, who were eager to share them with their families. At the end of the pilot study, when both experimental groups were asked to suggest a name for ChatGPT, the E1 group chose "Breathless Intelligence," while the E2 group chose "Fun Scholar".

During the text-generation process, the initial prompts were determined by the students, and the texts were generated accordingly. The texts were then read and critically evaluated by the students in terms of their consistency with the initial prompts. Certain sections that the students had not anticipated but found engaging prompted them to create new prompts. In this way, the texts were produced through a critical and creative approach.

For students to engage more deeply with the materials, the prompts need to be clear, concise, and focused (Huang et al., 2023). An effective ChatGPT prompt should be specific, applicable, and capable of attracting the attention of the target audience (Nazari & Saadi, 2024). The prompts were created in accordance with these principles and the students' requests. The texts, visuals, and comprehension questions were developed using the following prompts: We are 4th-grade primary school students. We want to learn about the properties of the Sun, Earth, and Moon. "Can you explain this in a fun and easy-to-understand way?" and "Can you ask questions about the story that require students to use their analytical and synthesis skills?"

The experimental procedure was implemented over 10 weeks, with four class hours per week in each group. In total, the procedure covered 80 class hours, with 40 h in each group. Each week, during two class periods, texts, comprehension questions, and visuals were generated based on the students' prompts. The following day, the text was printed in color, copied for all students, and used during two class periods. Examples of the generated texts are presented below.

An excerpt from the text The Fairy Tale Forest, whose main idea is "if we are together, we can solve many problems", is presented below:

> *Once upon a time, in the Fairy Tale Forest, there was a world filled with colorful flowers and magical creatures. Princess Berrin of the Fairy Tale Forest had been captured by the wicked witch, Kötüselda. While everyone was searching for a way to rescue the princess, a kind rabbit named Pamuk, who lived deep in the forest, 1 day discovered an ancient book. The book was a magical guide that would help trace the path of a lost fairy tale.*

An excerpt from the text The Day of Farewell, whose theme is "the students' graduation ceremony", is presented below:

> *Once upon a time, there was a charming school with a colorful garden, a place where every child longed to study—Vehbi Koç Primary School. In the schoolyard stood a grand plane tree stretching toward the sky. On one of the most beautiful mornings of spring, the students of Class 4/M rushed excitedly beneath this tree, preparing for their graduation ceremony. The sun was shining, the birds were chirping, and the air was filled with the fragrance of flowers. Yet, while enjoying this beautiful morning, the students' faces reflected both joy and sorrow, for these were among their final days of bidding farewell to their school.*

After completing the experimental procedure, the post-tests were administered to both the experimental and control groups. Because some students were absent during these procedures, the schools were revisited to ensure their participation. The administration of the post-tests lasted 2 weeks. The study was completed in 14 weeks, including the pre-test, pilot study, experimental procedure, and post-test.

In the use of artificial intelligence, ethical concerns such as family life, social and moral values, and the protection of personal data should be given heightened consideration in the case of primary school students. The Ministry of National Education (2025) supports the use of artificial intelligence provided that it complies with ethical guidelines and ensures the protection of student data. In practice, secure internet access provided by the Ministry was employed to guarantee online safety. Since the intervention was implemented in the classroom under the



Fig. 2 Sample photographs from the implementation.

supervision of the researcher, all necessary safety measures were taken by the researcher. Teachers must also take precautions to ensure that children use artificial intelligence safely. Furthermore, by allowing students to explore and become familiar with AI, teachers can facilitate safe and meaningful interaction. For this purpose, Williams, Park, and Breazeal (2019) recommend the PopBots platform, through which students create their own artificial intelligence tools. Such interactions enable students to recognize AI technologies and foster awareness and responsibility toward artificial intelligence.

Examples from the experimental implementation are presented below in Fig. 2.

Limitations. This study has several limitations Students were allowed to use their final class activity hours for the experimental application. During these hours, students showed signs of cognitive and emotional fatigue. Additionally, the E1 group's classroom being on the garden floor and the presence of physical education lessons in the schoolyard during the last class hours made the implementation more challenging. In this respect, different challenges were encountered in classroom management across groups. Due to the crowded nature of the classes, the chosen seating arrangement made it difficult for students to read the ChatGPT-generated texts from the smart board. Therefore, the texts were enlarged through an additional process to facilitate reading. Furthermore, several limitations should be acknowledged: potential biases stemming from the teacher factor, the difficulty of ensuring consistent implementation across classrooms, the presence of students' personal technological devices, parental influence, students' access to AI tools outside of school, the risks that may arise when students use AI independently, and the inability to fully control students' interest in these tools. Another limitation concerns the transferability of the research findings to certain cultural and linguistic contexts.

Findings

Quantitative findings of the study. The results of the Mann–Whitney *U*-Test comparing the number of the pre-test reading errors of the E2 and C1 groups from the study groups are presented below in Table 4.

Table 4 Mann-Whitney U-Test Results for Pre-Test Reading Error Counts of Groups E2 and C1.

Group	*n*	Order Mean	Order Total	*U*	*p*
E2	25	29.24	731	319	449
C1	29	26	754		

$p < 0.05$.

Table 5 Independent samples *t*-test results for pre-test comprehension and motivation scores of groups E2 and C1.

	Group	*n*	$\bar{x}$	S	sd	*t*	*p*
Comprehension Scores	E2	25	7.6	3.81	52	312	756
	C1	29	7.27	3.78			
Motivation Scores	E2	25	68.48	8.08	52	028	977
	C1	29	68.41	8.88			

$p < 0.05$.

Table 4 shows no statistically significant difference in the number of pre-test reading errors between the E2 and C1 groups ($U = 319$).

The results of the Independent Samples *t*-test comparing the E2 and C1 groups in terms of pre-test comprehension and motivation scores are presented in Table 5.

As shown in Table 5 no statistically significant differences were found between the E2 and C1 groups in pre-test comprehension and motivation scores [$t(52) = 0.312$; $t(52) = 0.028$].

ANOVA results comparing the study groups on post-test reading errors, comprehension, and motivation scores are presented in Table 6 below.

As shown in Table 6, statistically significant differences favoring the E1 and E2 groups were found in post-test reading errors, comprehension, and motivation scores [$F_{(3)} = 13.695$, $p < 0.05$; $F_{(3)} = 22.44$, $p < 0.05$; $F_{(3)} = 6.42$, $p < 0.05$]. Based on these findings, it can be concluded that, when the groups were compared, the experimental intervention statistically significantly reduced the number of reading errors and increased comprehension and motivation scores in favor of the experimental groups.

To calculate the effect sizes of the significant differences, the *eta squared* (η^2) value was computed for each dependent variable. Richardson (2011) proposed the following formula for calculating eta squared. Cohen (1988) established the following guidelines for interpreting the values obtained using this method: 0.0099 represents a small effect, 0.0588 represents a medium effect, and 0.1379 represents a large effect.

$$\eta^2 = \frac{\text{Sum of Squares(Between Groups)}}{\text{Sum of Squares(Total)}}$$

The formula above has been formulated below to calculate the η^2 values for all dependent variables.

$$\eta^2(\text{reading motivation}) = \frac{SS(1239.723)}{SS(7738.762)} = 0.16\eta^2(\text{reading error})$$
$$= \frac{SS(891.823)}{SS(3084.233)} = 0.289$$

$$\eta^2(\text{understanding score}) = \frac{SS(534.587)}{SS(1336.533)} = 0.399$$

As a result of the above procedures, when the values obtained for reading motivation ($\eta^2 = 0.16$), reading error count ($\eta^2 = 0,289$), and comprehension score ($\eta^2 = 0.399$) are evaluated according to

Table 6 ANOVA results of the post-test scores of the experimental and control groups on reading errors, comprehension and motivation.

	Group	*n*	$\bar{x}$	S	sd	*F*	*p*	Significant difference
Number of reading errors	E1	27	4.14	4.07	3	13.695	000	E1-C1
	E2	25	3.34	2.89				E1-C2
	C1	29	10.22	5.47				E2-C1
	C2	24	8.45	5.59				E2-C2
Comprehension scores	E1	27	10.37	3.46	3	22.44	000	E1-C1
	E2	25	11.92	2.81				E1-C2
	C1	29	7.24	2.48				E2-C1
	C2	24	6.25	2.28				E2-C2
Motivation scores	E1	27	72.85	5.38	3	6.42	001	E1-C1
	E2	25	73.28	7.61				E1-C2
	C1	29	66.79	9.34				E2-C1
	C2	24	65.58	9.11				E2-C2

$p < 0.05$.

Table 7 Student opinions on the effects of the ChatGPT-4 intervention on language skills.

Effects on the formal aspects of reading		***f***
Enrichment of vocabulary	E1S6, E1S7, E1S8, E1S10, E1S15, E1S18, E1S22, E1S23, E1S25, E2S1, E2S2, E2S3, E2S7, E2S12, E2S16, E2S18, E2S23	17
Improvement of reading comprehension	E1S7, E1S6, E1S22, E1S25, E2S7, E2S17	6
Improvement of reading fluency	E1S15, E1S25, E2S3, E2S23, E2S24	5
Improvement of inferring word meanings from context	E1S8, E1S25, E2S7	3
To better understand the texts in which he/she is the protagonist	E2S7, E2S12, E2S23	3
Improvement of identifying with the characters while reading	E2S16	1
Improvement of the ability to identify the main idea	E1S23	1
Effects on the affective aspects of reading		*f*
Increase in reading motivation	E1S6, E1S8, E1S18, E2S1, E2S17, E2S18, E2S23	7
Increase in curiosity about the end of the reading text	E1S6, E1S15, E2S7,	3
Overcoming excitement while reading	E1S8, E2S16, E2S23,	3
Reading with enjoyment	E1S5, E1S7	2
Overcoming shyness while reading	E2S3, E2S24	2
Improvement of imagination through reading	E1S6, E2S7	2
Reduction of reading anxiety	E2S16	1
Reduction of anxiety about misreading	E2S24	1
Increased interest in books similar to the content of the texts	E2S23	1
Ability to recognize that there can be very beautiful stories	E1S10	1
Effects on the social aspects of reading		
Sharing the texts with family members	E2S12, E2S23	2
Having friends notice that i am a good reader	E1S23	1
Improving relationships with friends	E1S10	1
Effects on speaking and writing skills		
Improvement of oral expression skills	E2S3	1
Ability to express oneself better as a result of improved reading skills	E1S18	1
Starting to write stories	E2S7	1

Cohen's (1988) criteria, it is observed that the effect of the significant differences for all variables is at a high level.

Qualitative findings of the study. Table 7 presents student opinions on the effects of the ChatGPT-4 intervention on language skills.

As shown in Table 7, the effects of ChatGPT-4 on language skills are categorized into four themes: effects on the formal aspects of reading, affective aspects of reading, social aspects of reading, and effects on speaking and writing skills. Seventeen students reported improvements in vocabulary; seven indicated an increase in reading motivation; six noted gains in reading comprehension; five mentioned enhanced reading fluency; and one reported getting along better with friends. Student E1S10 stated: *"It's really great. I get along well with my friends now. It's very good—we created a text together, and I never mess it up. I get along with them in a healthy way."* Student E2S3 expressed: *"I used to get excited on Tuesdays and Fridays when you came. The night before activity days, I would read books to prepare. I also used to be shy around my friends, but that's over now. For example —how should I put it… when I read a book at home, I couldn't really explain it to my mom; I felt a bit shy about that. But once, I read a 200-page book and explained it to her nicely."*

Table 8 presents student opinions on the characteristics of the texts created with ChatGPT-4.

When Table 8 is examined, the features of the texts created with ChatGPT-4 are grouped under two themes: language and

Table 8 Student opinions on the features of texts generated with ChatGPT-4.

Characteristics of the texts in terms of language and expression		f
Texts described as fun, engaging, and adventurous	E1S5, E1S6, E1S7, E1S8, E1S10, E1S15, E1S18, E1S22, E1S23, E1S25, E2S1, E2S12	12
Texts contributing to knowledge acquisition	E1S8, E1S10, E1S15, E2S1, E2S24	5
Improvement in reading and comprehension during Turkish lessons	E1S10, E1S15, E1S18, E1S23, E2S18	5
Texts increasing students' desire to read	E1S6, E1S7, E1S15, E2S16	4
Texts being creative	E1S5, E2S2, E2S16	3
Texts arousing curiosity	E1S6, E1S8, E1S10	3
Supernatural beings and events in the texts attracting attention	E1S6, E1S7, E1S22	3
Texts providing the opportunity to realize dreams	E1S10, E2S7	2
Increasing attention to Turkish lesson activities	E2S7, E2S23	2
Texts enabling students to recognize and develop their creativity	E1S5	1
Characteristics of the visuals accompanying the texts		
Visuals compatible with the texts	E1S8, E1S22, E1S23, E2S1, E2S7, E2S12, E21S16, E2S23,	8
Visuals compatible with the texts only when generated with new prompts	E2S2, E2S17, E2S18	3
Vivid visuals	E2S3, E2S7	2
Visuals enhancing students' motivation to read	E2S7	1

Table 9 Student opinions on the adequacy of ChatGPT-4 in text generation.

Advanced features of ChatGPT-4		f
The sentences in the texts that are related and harmonious	E1S7, E1S8, E1S18, E1S22, E1S25, E2S1, E2S2, E2S3, E2S16, E2S17	10
ChatGPT-4 giving the meaning of words according to the context of the text	E1S25, E1S6, E1S7, E1S8, E1S15, E1S18, E2S16, E2S18	8
ChatGPT-4 giving the correct meaning of words	E2S1, E2S16	2
Easy access to the meaning of words in ChatGPT-4	E2S17	1
Aspects of ChatGPT-4 that need to be improved		
ChatGPT-4 should be able to convert text into video.	E1S5, E1S6, E1S18, E1S25, E2S1, E2S7, E2S12, E2S16	8
There should be no gaps between the sentences in the text, even if they can be corrected with new prompts.	E1S6, E1S15, E2S7, E2S24	4
Real photos should be usable in the visuals.	E1S25, E2S17, E2S23, E2S24	4
It should add more emotion to the texts.	E2S1, E2S2	2
The texts should be longer.	E1S7, E1S8	2
There should be no similarities between the texts	E1S15	1
Animations related to the text should be created	E2S7	1
Different writing styles should be available	E2S7	1
ChatGPT-4 should avoid using inverted sentences.	E2S16	1
There should be more conversations in the texts	E2S18	1
It should be able to create music related to the texts.	E2S17	1

expression, and visuals. Twelve students stated that the texts were entertaining, engaging, and adventurous; eight reported that the visuals matched the text; five noted that the texts increased their knowledge; and three mentioned that the visuals became consistent only after new prompts. The student with the code E1S15 stated, "I read better than the books we read in class. The fact that it is interesting and action-packed caught my attention. Since we started it in the first week, my reading has improved a bit. My mom and dad always say so when I read to them. It really helped me read well." The student with the code E2S16 stated, "My vocabulary has improved. I didn't really like reading before; I used to set a page limit, read up to that page, and then stop. Now I say I'll read up to 40 pages, but I often end up reading more. My vocabulary and interpretation have improved. I immerse myself in the story, and while reading, I feel as if I am living it myself. This makes me want to continue the story and keeps me curious about what will happen next".

Student opinions regarding the adequacy of ChatGPT-4 in text generation are presented in Table 9.

When Table 9 is examined, the adequacy of ChatGPT-4 in text generation is categorized under two themes: its advanced features and the areas needing improvement. Ten students stated that the sentences in the texts were coherent; eight mentioned that ChatGPT-4 provided word meanings according to context; and eight suggested that ChatGPT-4 should be able to convert texts into videos. The student with the code E2S16 stated, *"For example, we mentioned that sometimes he constructs inverted sentences; he could correct those, he could improve his comprehension more."* The student with the code E2S18 stated, "Some texts turned out exactly as I wanted; in some, there was very little dialogue; and in others, there was less than I expected".

Discussion and conclusion

In the study examining the effects of texts created with students' prompts using ChatGPT-4 on reading, comprehension, and reading motivation, significant differences were found in favor of the experimental groups for all variables. As a result of the study, the comprehension scores of the experimental groups improved to a statistically significant degree compared with the control groups. In the qualitative research group, six students (E1S7, E1S6, E1S22, E1S25, E2S7, E2S17) reported that their comprehension skills had improved. Similarly, in a study conducted by Gharaibeh and Basulayyim (2025) with primary school students experiencing reading and comprehension difficulties, it was found that the comprehension scores of the experimental group significantly increased compared with those of the control group

when supported by ChatGPT. In another study, Al-Otaibi and Al-Homidhi (2025) examined the effects of ChatGPT on second-language comprehension skills and determined that the experimental group demonstrated notable development in both creative and critical comprehension abilities. In a meta-analysis of 51 studies conducted between 2022 and 2025, the effectiveness of ChatGPT in learning environments was examined, and it was determined that ChatGPT had positive effects on students' learning, perception, and higher-order thinking skills (Wang & Fan, 2025). In the qualitative research group, three students (E1S5, E2S2, E2S16) stated that the texts were creative; three students (E1S6, E1S8, E1S10) indicated that the texts increased their sense of curiosity; one student (E1S5) expressed that the texts helped them recognize their own creativity; and two students (E1S6, E2S7) reported that their imagination improved. When the quantitative and qualitative findings of the study are considered together with the existing literature, it can be concluded that the ChatGPT-supported intervention enhanced students' comprehension skills in a way that also encompassed variables such as curiosity, imagination, and creativity.

An important contribution of this study is that it integrates artificial intelligence applications with instructional methods. In Wang and Feng's (2024) study on the impact of ChatGPT on improving foreign language reading and comprehension skills, the ChatGPT-supported learning process was compared with the paper-based process. Comprehension in the paper-based learning group was around 63%, whereas in the ChatGPT-supported group it exceeded 86%. In the study by Achour, Laanoui, and Ourahay (2024), personalized interaction between ChatGPT and students supported their understanding of complex texts. However, it was emphasized that for comprehensive development of comprehension skills, ChatGPT applications should be supported by traditional teaching methods rather than replace them. In the current study, although ChatGPT-4 was used to create texts and activities, the texts were printed on paper and the reading and comprehension activities were carried out through traditional teaching methods. In the qualitative research group, three students (E1S8, E2S16, E2S23) reported overcoming their excitement about reading; two students (E2S3, E2S24) indicated that they were freed from the effects of reading-related shyness; one student (E2S24) stated that their anxiety about misreading decreased; one student (E1S10) noted improved relationships with peers; one student (E2S3) reported progress in oral expression skills; and one student (E1S23) shared that their peers had recognized them as a good reader. When considered alongside findings from the literature, the views of participating students suggest that the interaction-based learning methods adopted in this study contributed to students' social and personal development. In this respect, combining technology with pedagogical approaches rather than using technology in isolation can be regarded as a more effective choice for fostering student development.

In Wang and Feng's (2024) study, 59.52% of students reading from paper reported being unable to complete the task due to limited vocabulary, whereas students using ChatGPT completed the tasks and understood the texts better because they received real-time vocabulary support. In Al-Siyabi's (2024) study with 5th-grade students, stories created and illustrated by ChatGPT enriched students' vocabulary by helping them learn words applicable to daily life. Furthermore, 17 students in the research group (E1S6, E1S7, E1S8, E1S10, E1S15, E1S18, E1S22, E1S23, E1S25, E2S1, E2S2, E2S3, E2S7, E2S12, E2S16, E2S18, E2S23) reported that their vocabulary had expanded—a finding consistent with the results reported in the literature. During reading comprehension tasks, ChatGPT offers students opportunities to access necessary information, provides personalized learning experiences, and enriches interaction with the text (Quiroz-Martinez et al., 2024). During the study, students used ChatGPT to learn the meanings of unfamiliar words in the texts, tailored to their personal needs, thereby enriching interaction between reader and text. This enabled students to take responsibility for their own understanding during the comprehension process. The fact that three students in the qualitative research group (E1S8, E1S25, E2S7) stated that they learned the meanings of unfamiliar words independently by inferring from the context of the text confirms this finding. Moon, Wold, and Francom (2017) aimed to improve the comprehension performance of 5th-grade students by designing iPad-based interactive activities that engaged them with texts and tasks, ensuring active participation in the reading comprehension process. Their study revealed a statistically significant improvement in the students' final test scores. Moreover, the students reported enjoying the activities more than traditional reading. The findings of this study, which emphasize active student participation and contribute to enriching the interaction between reader and text, can be considered indicators of the study's alignment with the literature.

In Afkarin and Asmara's (2024) study, the effect of ChatGPT on students' motivation and academic success was examined. Students in the ChatGPT group reported higher motivation. Rai, Shapsough, and Zualkernan's (2024) study examined ChatGPT-4-generated stories in terms of fluency and logical consistency. They noted that student feedback would provide an opportunity to create texts that better meet learners' needs and preferences. Because AI-supported texts meet students' needs and support their aspirations, they contribute to the development of comprehension skills (He, 2024). The statement of the student with the code E2S7 *"… I mean, honestly, it affected me because we are included in that text, in a text that includes us, it tells our imagination, our story. For example, I didn't really believe in imagination, like when writing a story, I thought, "Stop writing it, it's just imagination, it doesn't become real. But these have changed me a bit more."* indicates that the approach mentioned in the literature has been realized. The fact that three students (E2S7, E2S12, E2S23) reported understanding texts better when they were the protagonists, and that two students (E1S10, E2S7) mentioned finding opportunities to realize their dreams through the texts, indicates that the texts were created in line with the students' needs and interests. This may have contributed to the development of students' reading motivation.

Bulut and Yildirim-Erbasli (2022) examined the ability of advanced language models, including ChatGPT, to generate texts and tasks that assess whether students comprehend the text. The study concluded that these models could design activities to assess students' comprehension, but it also emphasized the limitation that they often produce stories with repetitive wording. The student coded E1S15 stated, *"ChatGPT got a lot of things wrong. For example, we created a story and asked for it to take place on Mars, but it reminded us of Candy World (one of the previously created texts). It took us back to Candy World—the stories were similar."* This highlights the weakness of ChatGPT-4 noted in the literature.

An important issue that needs to be addressed is the practical implications of the effect sizes obtained in terms of reading errors, comprehension scores, and reading motivation. Effect size is considered an indicator of the practical significance of results beyond statistical significance (Özsoy & Özsoy, 2013). Bus, Shang, and Roskos (2024) examined studies conducted between 1970 and 2020 within the scope of independent reading practices outside of traditional classroom reading activities, and reported an overall significant effect size for word recognition, decoding, fluent reading, and reading attitudes. Similarly, Alqahtani (2024), in a meta-analysis of studies involving students with reading

difficulties, found that technology-based intervention programs had a substantial effect size. It can be stated that this intervention, which yielded results consistent with the literature, offers a strong model that teachers can employ to support students' reading skills.

In this research group, the texts generated by ChatGPT based on students' prompts enhanced their reading, comprehension, and reading motivation. This outcome can be integrated with multiple learning objectives included in curricula, such as vocabulary enrichment, making inferences, and identifying the main idea and theme. The widespread availability of internet access in schools, along with the use of computers, projectors, and smart boards, can facilitate the integration of student-generated prompts with curricular objectives. In this way, artificial intelligence can be incorporated both in language classes and during free reading periods, thereby fostering the development of students' reading and comprehension skills.

The following recommendations are based on the study's findings.

(i) Teachers can support students' reading skills during free reading sessions by using ChatGPT to generate texts and visuals aligned with their prompts.

(ii) When teachers inform parents about ChatGPT's contributions, it can help enhance children's reading and comprehension skills at home.

(iii) The study included all students in the classes that formed the experimental groups. To more clearly reveal ChatGPT's effect on reading and comprehension, future studies could focus on groups consisting solely of students who struggle with these skills.

(iv) Since some students indicated improvements in their speaking and writing skills, further studies could examine ChatGPT's impact on these areas.

(v) Teachers can utilize ChatGPT to guide students in creating their own storybooks.

(vi) Comparative studies can be conducted to examine the effects of ChatGPT and other artificial intelligence models on students' reading, comprehension, and reading motivation.

(vii) Longitudinal studies can be carried out to investigate the sustainability of ChatGPT's effects on students' reading, comprehension, and motivation.

(viii) Teachers may also benefit from ChatGPT in areas related to reading skills included in the curriculum, such as developing daily lesson plans, designing activities, and preparing prediction, interpretation, and inference questions.

Data availability

The data collected during the research process has been archived. This data can be shared with the relevant parties upon request.

Received: 16 February 2025; Accepted: 10 November 2025;
Published online: 02 December 2025

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Acknowledgements

I would like to thank the principals of the schools where I conducted this research, Ahmet ÖNAL and Ferman KARATAŞ; and the primary school teachers, Emel KAPLAN, Sacide IŞIK, Sevcan KAYIHAN and Fatih GİRGİN. This study was not funded by any institution or organization.

Author contributions

"All stages of this research were carried out by YB, the sole author of the article."

Competing interests

The author declares no competing interests.

Ethical approval

Ethical approval for this research was obtained from the Research and Publication Ethics Committee of Bursa Uludağ University, with the decision dated February 23, 2024, and numbered 2024-02.

Informed consent

Written informed consent indicating the voluntary participation of the students in the study was obtained from the parents of the students in the research group on February 26, 2024. The research was conducted in accordance with the *Research Application Permit Circular* issued by the Ministry of National Education, which sets out the principles for conducting studies with students.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1057/s41599-025-06265-x.

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